

Mapping basal ganglia circuitry in altered states of consciousness: from pathology to pharmacology

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Abstract

The precise neuromodulatory mechanisms underlying consciousness and its disorders, despite growing evidence in both animals and humans, remain poorly understood at the subcortical level. The basal ganglia, a key component of the cortico-basal ganglia-thalamo-cortical loop, are known to play a crucial role in mediating consciousness and behavioural responsiveness to the environment. Here, we aimed to present a comprehensive mapping of the distinct contributions of basal ganglia nuclei and their associated neurotransmitter systems in pharmacologically induced and pathological loss of responsiveness.

We used functional MRI to provide a systematic investigation of 9 major nuclei (putamen, caudate, nucleus accumbens, globus pallidus external, globus pallidus internal, substantia nigra pars compacta, substantia nigra pars reticulata, ventral pallidus, and subthalamic nucleus) and their functional relationship with 19 major neurotransmitter systems in loss of responsiveness, in both healthy volunteers under propofol anaesthesia (n=16) and patients with disorders of consciousness (DOC) (i.e. minimally conscious, unresponsive wakefulness syndrome) (n=22).

We found that in pharmacologically-induced loss of responsiveness, the functional connectivity changes of the putamen (Pu) correlated with the norepinephrine transporter (NET), and the globus pallidus external (GPe) with the serotonin transporter (5-HTT) and vesicular acetylcholine transporter (VACHT); additionally, altered connectivity in the ventral pallidum (VeP) was associated with the norepinephrine transporter (NET). In contrast, with pathological-induced loss of responsiveness, we found that connectivity changes of the caudate were associated with the

serotonin transporter (5-HTT), dopamine transporter (DAT), and GABA_A receptor, and connectivity changes of the substantia nigra pars compacta (SNc) were associated with the cannabinoid (CB1) receptor. Finally, we report that among all basal ganglia nuclei, a key dopamine-rich area of the basal ganglia — the substantia nigra — was found to have the largest DOC subgroup difference in functional connectivity, following stratification based on mental-imagery task responsiveness. Critically, we provide evidence that pharmacological and pathological loss of responsiveness involve distinct neurotransmitter system contributions across individual basal ganglia nuclei, offering a novel framework for targeted therapeutic interventions in patients with disorders of consciousness.

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Introduction

Conscious experience arises from interactions between cortex, basal ganglia, thalamus, and cerebellum.^{1–3} The basal ganglia are part of a ‘cortico-basal ganglia-thalamo-cortical loop’, in both animals^{4–7} and humans.^{8,9} Understanding this circuitry’s functional organisation is fundamental to cognition and consciousness; recent work explored its role in anaesthetised humans¹⁰ and post-coma recovery following brain injury.¹¹

There also remains a gap in understanding how neurotransmitter systems modulate altered states of consciousness.¹² To address this, we map the functional roles of individual basal ganglia nuclei, and their neuromodulatory interplay, in pharmacologically and pathologically perturbed consciousness, indicated by loss of behavioural responsiveness to the environment. To achieve this, we systematically investigate distinct nuclei and their functional relationship to cortical and subcortical areas using neuroimaging datasets from (i) healthy propofol-anaesthetised volunteers (n=16) and (ii) patients with disorders of consciousness (DOC) (n=22), with n=13 in minimally conscious state (MCS) and n=9 in unresponsive wakefulness syndrome (UWS). With our exploratory approach, we analyse DOC as a cohort while acknowledging MCS-UWS distinctions and “graded” awareness, further stratifying patients by neuroimaging evidence of covert consciousness¹³ to capture clinical nuances and subtle functional changes. Anaesthesia is a reversible, controllable experimental surrogate for the behavioural unresponsiveness (even in the presence of stimuli such as shaking or loud noises) that characterises DOC.^{2,14–19} Anaesthesia mirrors the behavioural unresponsiveness of DOC patients while also reproducing key neurophysiological features i.e. constrained repertoire of brain dynamics and increased structure–function coupling^{15,20}. It also affords high degree of experimental control for reproducible studies of consciousness, where behavioural responsiveness is typically used as an (albeit imperfect) indicator^{15,20} – although it should be noted that most anaesthetics (including propofol) suppress both behavioural responsiveness and physiological arousal, whereas spontaneous cycles of higher and lower physiological arousal can be observed in DOC patients, despite persistent unresponsiveness to the environment (hence the diagnosis ‘unresponsive wakefulness’ to highlight this dissociation).

We thus probe basal ganglia contributions to behavioural responsiveness to the environment (the most common indicator of consciousness in the clinic) and their relationship to macroscopic hierarchical functional organisation and neurotransmitter systems. Beyond refining our neurochemical understanding of this circuitry, this work has therapeutic implications for targeted neuromodulation, given uncertainty over optimal intervention sites^{12,21} and variable outcomes in DOC.²²

Materials and methods

Data acquisition

Healthy awake and anaesthetised volunteers: “Western Dataset”

The anaesthesia dataset was collected between May and November 2014 at Robarts Research Institute in London, Ontario, Canada, received approval from Western University Ethics board²³, and has been previously published.^{16,19,24,25} Details of anaesthetic procedure and MRI acquisition are in Supplementary Materials ‘Methods’.

DOC Dataset

MRI data for 24 DOC patients were collected at the Wolfson Brain Imaging Centre (WBIC) Addenbrooke’s in Cambridge, UK, between January 2010 and July 2015; the DOC dataset employed in our study has been published before.^{16,17,19,25} Demographics are provided in Table 1. Details of patient selection and assessment are in Supplementary Material (‘Methods’).

Healthy awake volunteers: “Cambridge Dataset”

We included an additional healthy awake control dataset to repeat and replicate our *DOC versus controls* analysis; this allowed us to run the DOC group-level analysis with a control dataset collected on the same scanner as the DOC patient dataset, with identical scanner acquisition parameters. Details of MRI acquisition are in Supplementary Material (‘Methods’).

Demographics are provided in Table 2. With respect to descriptive statistics, the DOC cohort included 22 patients (14 males, 8 females; mean age = 37.6 ± 16.0 years, SD = 16.0, range 17-70). The “Western” dataset included 16 healthy volunteers (12 males, 4 females; age range 18-40 years). Gender distribution did not significantly differ between the “Western” and DOC groups after chi-squared test ($\chi^2(1) = 0.90$, $P = 0.34$); a statistical test of age differences could not be conducted because only age range was available for this dataset. The “Cambridge” dataset included 17 healthy volunteers (8 males, 9 females; mean age = 35 years; age range 19–53). Gender distribution did not significantly differ between the “Cambridge” and DOC groups after chi-squared test ($\chi^2(1) = 1.11$, $P = 0.29$), and age was also not significantly different between groups (Welch’s two sample t-test $t(32.16) = -0.77$, $P = 0.45$).

Data pre-processing

Functional and structural images were pre-processed using SPM12 (<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>). Details of pre-processing are in Supplementary Material (‘Methods’).

Basal ganglia parcellation atlas

We implemented seed-based functional connectivity (FC) analysis to identify the target brain regions that functionally interact with individual basal ganglia nuclei. Seeds (nuclei) are defined as regions-of-interest (ROIs).²⁶ There are few digital atlases for the basal ganglia.^{27–30} We chose a recent high-resolution probabilistic *in vivo* anatomical atlas with fine parcellation of subcortical nuclei, capable of discerning between specific basal ganglia subdivisions, developed from 168 adults in the Human Connectome Project.²⁹ Therefore, our pipeline does not rely on subject-level grey matter segmentation to identify subcortical nuclei; rather, we employed a standard high-resolution probabilistic atlas²⁹ in MNI space. We performed co-registration of basal ganglia masks into the pre-processed functional image space of the healthy volunteers and patients, utilising nearest neighbour interpolation. This enabled us to apply the atlas-defined ROIs and conduct our analysis in a common space across all patients and controls. We calculated that resulting masks

contained the following number of voxels for the ROIs: SNc (45 voxels); SNr (93 voxels); GPe (281 voxels); GPi (145 voxels); Ca (1617 voxels); Pu (1802 voxels); NAC (130 voxels); VeP (35 voxels); STH (52 voxels). While partial volume effects and anatomical variability may pose challenges, particularly in brain injury patients, our analyses focused on group-level statistical maps. This enabled a consistent application of atlas-defined ROIs across patients and controls, while mitigating individual anatomical variation through group-level analyses. Furthermore, our stringent quality control and anatomically validated high-resolution atlas, with more accurate external boundaries delineating than current available histology-based atlases,³⁹ as well as our rigorous spatial normalisation, is therefore consistent with current methodological standards and enhance the robustness of our group-level findings. Moreover, using 6 mm FWHM smoothing kernel level reflects a balance between spatial specificity and signal-to-noise optimisation, particularly for DOC¹⁹ and TBI,^{31,32} and has been commonly implemented in brainstem and basal ganglia fMRI studies.^{17,33,34}

FC analysis and statistics

We analysed the fMRI data of n=16 healthy controls under deep sedation (mean plasma propofol concentration of 2.68 μ g/mL) and n=22 DOC patients. Basal ganglia masks corresponded to 9 major nuclei (bilaterally): Pu (putamen), Ca (caudate), NAC (nucleus accumbens), GPe (globus pallidus external), GPi (globus pallidus internal), SNc (substantia nigra pars compacta), SNr (substantia nigra pars reticulata), VeP (ventral pallidus), STH (subthalamic nucleus).

FC was calculated using CONN (<https://web.conn-toolbox.org>) as seed-to-voxel analyses from 9 basal ganglia nuclei to examine their functional interactions with the rest of the brain. Temporal correlations for basal ganglia seeds were computed for all other voxels in the brain using a general linear model (GLM). For the anaesthesia dataset with a within-subject experimental design, FC analyses were initially performed individually, then individual seed-to-voxel parameter estimate images were entered into group-level analyses. Paired-sample *t* tests were used for differences in the participants among the condition of deep sedation versus awake. For the DOC dataset with a between-subject experimental design, two-sample *t*-test (DOC vs. control) was performed at group level. We compared the DOC group to healthy awake “Western” controls from the anaesthesia experiment, and also compared the DOC group to the “Cambridge” controls. The group-level

inference of both datasets was corrected for multiple comparisons using random field theory [voxel level threshold of $p < 0.005$ (uncorrected) and cluster level $p < 0.05$ (FWE-corrected for multiple comparisons)].

Radarplots were additionally computed to indicate the intrinsic connectivity network (ICN) spatial involvement (ICNi) in contrasts of deep sedation vs. awake and DOC vs. control. These plots display the voxel overlap between FC results and canonical ICNs; these ICNs were defined by an atlas³⁵ including 10 resting-state networks (RSN) from the “ICN_atlas” toolbox (https://www.nitrc.org/projects/icn_atlas/):³⁶ VN 1, 2, and 3 (“visual”): medial, occipital pole, and lateral visual areas; DMN (“default-mode”): medial parietal (precuneus and posterior cingulate), bilateral inferior-lateral-parietal, and ventromedial frontal cortex; CN (“cerebellum”): cerebellum; SMN (“sensorimotor”): supplementary motor area, sensorimotor cortex, and secondary somatosensory cortex; AN (“auditory”): superior temporal gyrus, Heschl's gyrus, and posterior insular, and includes primary and association auditory cortices; ECN (“executive-control”): medial-frontal areas, including anterior cingulate and paracingulate; FPN1 and 2 (“fronto-parietal”): frontoparietal areas (only maps which are strongly lateralised).^{35,36} FPN1 corresponds strongly to perception-somesthesia-pain, and FPN2 to cognition-language paradigms, consistent with Broca's and Wernicke's areas.³⁵⁻³⁷

We also sought to test the behavioural relevance of FC changes within our DOC cohort. Patients were stratified into (i) negative fMRI responders to the Tennis Task (“task non-responders”, fMRI-) and (ii) positive fMRI responders (“task responders”, fMRI+). To investigate subtle FC changes within this DOC cohort, we compared the two DOC subgroups and computed a group-level contrast of *fMRI- versus fMRI+*. We computed unthresholded and thresholded cortical SPM-t maps that survived multiple comparisons using random field theory [voxel level threshold of $p < 0.005$ (uncorrected) and cluster level $p < 0.05$ (FWE-corrected for multiple comparisons)]. As a follow-up, to quantify the strength of these FC changes in individual patients, we computed a differential FC (dFC) score for each patient defined as the mean absolute voxel-wise difference between that patient's individual FC map and the DOC subgroup-level contrast map (fMRI- vs fMRI+), normalised by the spatial standard deviation of the contrast map. The grouped box plot displays

these dFC scores corresponding to each subgroup: (1) fMRI- or task non-responders' dFC from the group-level contrast, and (2) fMRI+ or task responders' dFC from the group-level contrast, across all basal ganglia nuclei. Lower dFC indicates closer alignment to the DOC subgroup contrast, whereas higher dFC indicates greater deviation. These scores were plotted in R and visualised in a box plot, where the y-axis represents dFC (standardised mean absolute difference) and x-axis depicts nuclei ("Basal Ganglia Nuclei"). Finally, a Welch two-sample t-test was applied to compare dFC values between fMRI- and fMRI+ subgroups (with FDR-correction to adjust for multiple comparisons across nuclei). *For additional robustness check, we performed Wilcoxon rank-sum test (with FDR-correction).* Mean dFC, median dFC, IQR, as well as raw p-values and FDR-corrected p-values for both tests are also presented (Supplementary Table 1).

PET neurotransmitter system analysis and statistics

To address our question of how FC changes induced by pharmacological (anaesthesia) and pathological (DOC) loss of behavioural responsiveness to the environment relate to neurotransmitter systems, we tested whether the voxel-wise cortical SPM-t maps from our group comparisons (contrasts of deep sedation vs. awake and DOC vs. control) correlated with 15 receptors and 4 transporters obtained from *in vivo* PET from healthy volunteers.³⁸ These included: dopamine (D1, D2, and DAT), noradrenaline (NAT), serotonin (5-HT1A, 5-HT1B, 5-HT2A, 5-HT4, 5-HT6, and 5-HTT), acetylcholine (α 4 β 2, M1, and VACHT), glutamate (mGluR5 and NMDA), GABA (GABA_A), histamine (H3), cannabinoid (CB1), and opioid (MOR) (Supplementary Table 2).³⁸

Both the PET and statistical maps of seed-to-voxel correlation t-scores were parcellated into 200 discrete functionally-defined cortical regions based on the local-global cortical functional atlas of Schaefer-200.³⁹ Significance was assessed against spin-based null models— where drug effects and neurotransmitters are spatially correlated with each other solely due to inherent spatial autocorrelation⁴⁰ — that preserve spatial autocorrelation of brain maps and therefore produce more accurate statistical estimates.^{41,42} Furthermore, all P-values were corrected for multiple comparisons using FDR-correction.

1

2 Results

3 We examined, in both pharmacological and pathological perturbations, i) which basal ganglia
 4 nuclei are *most* central to loss of behavioural responsiveness; ii) *how* these functional changes
 5 occur through neuromodulatory mechanisms involving major neurotransmitter systems in the
 6 brain; and tested iii) the behavioural relevance of these FC alterations in DOC patients. We
 7 analysed fMRI data of n=16 healthy controls under deep sedation (mean plasma propofol
 8 concentration of 2.68 $\mu\text{g/mL}$) and n=22 DOC patients (n=13 MCS and n=9 UWS), and used a
 9 high-resolution probabilistic *in vivo* atlas delineating subcortical nuclei complementing histology-
 10 based atlases.²⁹ We focused solely on basal ganglia nuclei.

11 Functional interactions of basal ganglia nuclei with whole brain in 12 anaesthetised healthy subjects

13 Firstly, we asked: *where* do FC alterations between basal ganglia nodes and cortex occur with
 14 pharmacologic loss of responsiveness in healthy volunteers? We thus investigated the spatial
 15 localisation of FC basal-ganglia-cortical changes with propofol-induced loss of responsiveness by
 16 computing whole brain connectivity (normalised as SPM-t) maps, unthresholded (Fig. 1A–I, *left*
 17 column) and thresholded (Fig. 1A–I, *middle* column), following seed-based FC analysis (contrast
 18 of deep sedation vs. awake). Thresholded statistical maps which survived corrections revealed that
 19 multiple basal ganglia nuclei altered their FC in loss of responsiveness induced by propofol. Pu,
 20 Ca, and VeP increased their FC with precuneus, caudate, and supplementary motor area,
 21 respectively, whereas STH decreased its FC with supramarginal gyrus. To establish cortical
 22 network involvement in SPM-t maps, we overlapped them with intrinsic brain connectivity
 23 networks (ICN) and calculated the involvement (ICNi) with a resting-state brain network atlas
 24 using the *ICN_atlas* toolbox.^{35,36} With pharmacological loss of responsiveness, we found
 25 differential cortical network FC patterns corresponding to distinct nuclei (Fig. 1A–I, *right*
 26 column). Among lower-order networks, there was a high involvement (ICNi) of VN for most
 27 nuclei, including Ca, NAC, GPe, GPi, SNc, SNr, VeP, STH; AUDN for Pu, Ca, GPe, GPi, SNc,
 28 SNr, VeP; SMN for the Pu, Ca, GPe, GPi, SNc, VeP; and CN for NAC, SNc, SNr. Alternatively,

for higher-order networks, there was a high involvement of FPN for Pu, Ca, NAC, GPe, GPi, VeP; ECN for Ca, NAC, GPe; and DMN for Pu, GPe, GPi, STH.

Secondly, we sought to identify the underlying neuromodulatory mechanisms that may account for these functional reconfigurations in the anaesthetised brain. Specifically, we address the following question how these propofol-induced FC changes significantly correlate — and negatively correlate — with 19 major neurotransmitter systems.³⁸ Hence, we demonstrate significant findings (Fig. 2) after statistical correction, derived from Schaefer's local-global parcellation atlas with 200 fine-grained cortical regions.³⁹ Significant positive correlations between propofol-induced FC changes and molecular targets were found for three basal ganglia nuclei. Thus, the nuclei's FC associated with neurotransmitter systems include: Pu with NET; GPe with 5-HTT and VACHT; and VeP with NET. Additionally, multiple nuclei's FC revealed significant negative correlations, notably: NAC with 5-HTT and DAT; SNc with A4B2 and MOR; SNr with A4B2, and STH with NET.

Functional interactions of basal ganglia nuclei with whole brain in DOC patients

Analogous to our previous experimental model, we asked: *where* do FC alterations between basal ganglia nodes and the cortex occur with pathological loss of behavioural responsiveness in DOC patients? We investigated the spatial localisation of FC basal-ganglia-cortical changes with pathologically-induced loss of responsiveness by computing whole brain connectivity (normalised as SPM-t) maps, both unthresholded (Fig. 3A–I, *left* column) and thresholded (Fig. 3A–I, *middle* column), following seed-based FC analysis (contrast of DOC vs. control), using the “*Western*” dataset. Thresholded statistical maps which survived corrections (voxel level threshold of $p < 0.005$ [uncorrected] and cluster level $p < 0.05$ [FWE-corrected for multiple comparisons]) revealed that multiple nuclei altered their FC in pathological loss of responsiveness. FC gains with frontal cortex were revealed by NAC, VeP, and Pu (also with anterior cingulate cortex). Additional FC gains involve GPe and STH with parahippocampal gyrus; GPi with middle inferior temporal gyrus; SNc gains with middle temporal gyrus; SNr with inferior temporal gyrus; Pu with lingual gyrus. Several nuclei revealed thalamic functional disconnections, including Ca, NAC, SNr, VeP, and STH. We also found differential cortical network FC patterns corresponding to distinct nuclei (Fig. 3A–I,

right column). Among lower-order networks, there was a strong involvement (ICNi) of VNs for Pu, Ca, NAC, GPe, GPi, SNc, SNr, VeP, STH; AUDN for Ca, SNr, SNc, VeP, STH; SMN for GPi, SNr, VeP; and CN for SNr, SNc, VeP. Alternatively, for higher-order networks, there was a strong involvement (ICNi) of FPN for Pu, NAC, GPe, GPi, SNc, SNr, VeP; ECN for Pu, Ca, NAC, GPe, GPi, SNc, SNr, VeP, and STH; and DMN for Pu, GPi, GPe, STH.

We further replicated our FC analysis with the “Cambridge” dataset, demonstrating that (i) cortical maps of FC changes resulting from DOC comparisons to each control dataset are significantly correlated, and (ii) highlighting similar supra-threshold clusters and (iii) exhibiting similar cortical network involvement (Supplementary Fig. 1).

Consequently, we examined the neuromodulatory mechanisms contributing to functional remodelling of the brain in DOC by addressing the question: how do these pathologically-induced FC changes significantly correlate with 19 major neurotransmitter systems?³⁸ We identified several basal ganglia nuclei for which the map of FC difference is significantly correlated with a neurotransmitter system PET map (even after controlling for spatial autocorrelation), and this correlation is significant in both comparisons (DOC vs “Western” controls, and DOC vs “Cambridge” controls) (Fig. 4). This approach complements the FDR-corrected analysis, controlling for potential false discoveries not by requiring a lower p-value, but by excluding any finding that is not replicable, either because it is only significant in one dataset, or because its direction (here, sign of the correlation) is not the same in both analyses. Thus, the rationale for this approach is that a true discovery should be not only significant, but also replicable. We have thereby confirmed the reliability of our findings by demonstrating that the FC maps for the two control datasets are similar, not only qualitatively (Fig. 3 “DOC vs. “Western” controls and Supplementary Fig. 1 “DOC vs. “Cambridge” controls) but also quantitatively (Fig. 4). Thus, in DOC, connectivity changes of the Ca significantly correlated with 5-HTT, DAT, and GABA_A, and connectivity changes of SNc significantly correlated with CB1. Additionally, several other nuclei’s connectivity changes significantly negatively correlated with neurotransmitter systems, including: SNr with A4B2, GABA_A, and mGluR5; SNc with GABA_A; GPi with GABA_A; STH with 5-HT1B; and VeP with GABA_A.

We implemented an in-scanner behavioural test^{13,43,44} which assessed the DOC patients’ level of covert consciousness, involving tasks of motor imagery (“Tennis task”) and spatial imagery

(“Navigation task”), while bypassing the need for over motor behaviour. DOC patients were further stratified into non-responders (fMRI-) and responders (fMRI+) to the Tennis Task. All task responders (n=7) were traumatic brain injury (TBI) patients (except for 1) and exclusively MCS. Among non-responders (n=15), 6 were MCS and 9 were UWS. This test has been demonstrated to serve as a proxy for consciousness, and adds value in providing nuanced cognitive evaluation alongside bedside assessment.^{13,43,45} Even healthy participants with intact cognitive processing can fail to respond, as demonstrated in investigations where 70-100% responded to task-based fMRI or EEG.⁴⁴ Thus, negative findings in these paradigms should be interpreted with caution.

We therefore tested the behavioural relevance of FC changes in patients by comparing the two DOC subgroups and computing a group-level contrast of *fMRI-* versus *fMRI+*. We computed unthresholded (Fig. 5A–I, left column) and thresholded cortical SPM-t maps (Fig. 5A–I, right column) that survived multiple comparisons using random field theory [voxel level threshold of $p < 0.005$ (uncorrected) and cluster level $p < 0.05$ (FWE-corrected for multiple comparisons)]. Six nuclei showed significant alterations in FC, notably: Pu with the caudate, pallidum, nucleus accumbens (FC increases) and precuneus, postcentral gyrus (FC decreases); Ca with the anterior cingulate, paracingulate gyrus, frontal operculum, central operculum (FC increases); NAC with the precuneus and superior parietal lobule (FC decreases); GPe with the precuneus (FC decreases); VeP with the posterior middle temporal gyrus and posterior superior medial temporal gyrus (FC increases); and STH with the right cuneal cortex and right occipital cortex (FC decreases).

Furthermore, we explored how the FC changes in the DOC subgroup comparison (*fMRI-* or *task non-responders* versus *fMRI+* or *task responders*) significantly correlated — and negatively correlated — with 19 major neurotransmitter systems. We present significant correlations for each nucleus between the cortical FC maps of the DOC subgroup comparison (contrast of *fMRI-* versus *fMRI+*) and 19 major neurotransmitter systems, that exhibited overlapping significance ($p_{spin} < 0.05$ after correcting for the role of spatial autocorrelation) in the same direction as the results obtained from comparing *DOC* versus “Western” controls and *DOC* versus “Cambridge” controls (Supplementary Fig. 3), where red represents significant positive correlations and blue represents significant negative correlations. Here, we found that in the [overlapping] DOC subgroup comparison, the only nuclei whose FC changes were associated with neurotransmitter

systems were the Ca, notably with 5-HTT, DAT, and GABA_A, and the SNc with CB1. Additionally, only GPi connectivity changes revealed significant negative correlations, with GABA_A.

Finally, we quantified the magnitude of these FC changes in individual patients by computing a differential FC (dFC) score for each patient, defined as the standardised mean absolute voxel-wise difference between that patient's individual FC map and the DOC subgroup contrast map (*fMRI- versus fMRI+*), normalised by the spatial standard deviation of the contrast map (Fig. 6). Lower dFC indicates closer alignment to the DOC subgroup contrast, whereas higher dFC indicates greater deviation. Thus, through this additional behavioural stratification, we identified which basal ganglia regions showed the greatest magnitude of FC changes with perturbations of consciousness across the spectrum of DOC pathology, thereby addressing our central hypothesis: which subcortical regions— basal ganglia nuclei— may be more relevant for allowing us to distinguish between 'conscious states'. For each nucleus, we applied Welch's two-sample t-test and performed post-hoc tests with FDR-correction to adjust for multiple comparisons. We found that multiple nuclei reached statistical significance: SNc (Welch's t-test FDR-corrected $P = 0.021$); GPi (Welch's t-test, FDR-corrected $P = 0.012$); SNr (Welch's t-test, FDR-corrected $P = 0.026$); and GPe (Welch's t-test, FDR-corrected $P = 0.010$). To account for the possibility of low sample size, heterogeneity of variances, and further validate the robustness and reliability of our findings, we additionally performed the Wilcoxon rank-sum test followed by FDR-correction, which produced significant consistent results: SNc (Wilcoxon rank-sum test, FDR-corrected $P = 0.020$), GPi (Wilcoxon rank-sum test, FDR-corrected $P = 0.023$), SNr (Wilcoxon rank-sum test, FDR-corrected $P = 0.042$), and GPe (Wilcoxon rank-sum test, FDR-corrected $P = 0.020$). Among these significant nuclei, the largest between-group difference in dFC was observed for SNc, followed by GPi, SNr, and GPe.

Discussion

In our study, we mapped the basal ganglia-cortical FC changes across pharmacological and pathological perturbations of consciousness (indicated by loss of behavioural responsiveness to

the environment) and linked them to major neurotransmitter systems. In anaesthesia, nuclei-specific FC changes correlated with NET (putamen, VeP), 5-HTT and VAcHT (GPe) map density. In contrast, in DOC, nuclei-specific FC changes correlated with the 5-HTT, DAT, and GABA_A (caudate) and CB1 (SNc) map density. It is noteworthy that many of the significant correlations involve transporters rather than receptors (3/4 in anaesthesia and 2/4 in DOC), with the serotonin transporter (5-HTT) being the only transporter map common to both conditions. Among all, the substantia nigra showed the largest between-group difference in the DOC subgroup comparison, highlighting a dopaminergic locus and its potential as a neuromodulatory target.

In healthy volunteers, we found increased FC between basal ganglia nuclei and distinct cortical regions with anaesthetic-induced loss of responsiveness, specifically from the putamen (a subcomponent of the striatum with major cortical inputs),^{46,47} which also has corticostriate projections to the precuneus,⁴⁸ a core region of the default mode network (DMN)^{49–51} that integrates both externally and internally driven information.^{48,52,53} The caudate (part of the striatum) increased its FC with areas of itself, which is in line with previous imaging studies demonstrating the anterior caudate's role in cognitive and emotional circuits, whereas the body or posterior areas were related to executive/action and perception-related networks.^{54,55} Furthermore, the GPi (a 'relay' nucleus receiving dorsal striatal input) contains predominantly GABAergic neurons in mice,⁵⁶ whereas the GPe contains sparse cholinergic neurons;⁵⁷ these nuclei are part of the broader 'dopaminergic system'.^{58,59} The GPe and putamen are directly/indirectly modulated by dopamine release and guide behaviour^{29,60,61} and the VeP drives activity in dopaminergic midbrain.^{29,62–64} Our findings *in vivo* extend this work by demonstrating propofol-related *nuclei-specific* FC changes associated with select neurotransmitter systems, notably with norepinephrine, serotonin, and vesicular acetylcholine transporters, providing evidence for altered neuromodulatory tone within striato-pallidal circuits in humans. Whilst *in vivo* dopaminergic VTA connectivity has been associated with network alterations in a propofol study,¹⁷ and basal ganglia nuclei have been previously linked to dopaminergic and cholinergic systems in conscious states,^{58,59} we outline a richer neuromodulatory basis for pharmacological loss of behavioural responsiveness to the environment (and possibly concomitant loss of consciousness), at the level of individual basal ganglia nuclei.

Furthermore, the nuclei with propofol-induced FC changes that correlated with neurotransmitter system maps were also coupled to lower-order networks (AUDN, SMN, and VN), with the latter linked to the GPe and VeP only, whereas higher-order involvement varied across the VeP (FPN), putamen (FPN and DMN), and GPe (FPN, DMN, ECN). Dynamic interactions between large-scale cortical networks are known to sustain consciousness.^{2,65} Additionally, subcortical regions functionally connect to networks modulating higher-order cognition, forming 'integrative

network hubs', which include the midbrain ventral tegmental (VTA), thalamic central lateral (CL) and parafascicular nuclei (Pf), pontomesencephalic tegmentum, and reticular and extrareticular arousal nuclei.⁶⁶ Brainstem arousal and thalamic intralaminar nuclei also functionally connect to DMN and SMN.⁶⁶ Our findings reinforce these inferences relevant to functional connectivity hubs, advancing a detailed subcortical framework supporting the organism's ability to behaviourally interact with the environment, and possibly consciousness more broadly.

In DOC patients, we found nuclei-specific FC changes correlated with neurotransmitter systems, notably between the caudate and 5-HTT/DAT/GABA_A, and the SNc and CB1, aligning with serotonergic^{67–71} and dopaminergic^{58,72–74} roles in basal ganglia circuits. Notably, the associations between nuclei-specific FC changes and neurotransmitter system targets replicated across the two independent control datasets ("Western" and "Cambridge") using identical pipelines and spin-tests, underscoring the robustness of our results. Dopamine is known to facilitate D1-mediated corticostriatal transmission ('direct' pathway) and suppresses D2-mediated transmission ('indirect' pathway),⁷⁵ reducing GPi/SNr activity and disinhibiting thalamo-cortical projection neurons critical for preparation and selection of behaviour.^{58,75} Behaviourally, ventral striatum dopamine influences reward-related stimulation on appetite and dorsal striatum dopamine influences sensorimotor coordination in consummatory responses.⁷⁶ Importantly, dopamine dysregulation underlies Parkinson's disease, with nigrostriatal dopaminergic neuronal loss depleting pallidal dopamine,⁷⁷ whereas hyperstimulation of striatum D2 receptors has been linked to psychosis, hallucinations, and schizophrenia.^{78–80} Furthermore, low extracellular GABA underpins tonic inhibition and conscious-state regulation, making them key targets for anaesthetics, and disruptions have been linked to epilepsy and Parkinson's.⁸¹ Specifically, both caudate and putamen have abundant GABA_A and D1-expressing MSNs for tonic conductance, and loss of dopaminergic drive in motor dysregulations (i.e. Parkinson's) elevate striatal GABA.⁸² We found FC changes of the SNc correlated with cannabinoid targets, aligning with known cannabinoid modulation of dopamine transmission;⁸³ specifically CB1 is found in the substantia nigra and VTA, embedding nigrostriatal and mesocorticolimbic dopaminergic neurons, and its dopamine modulation underlies neuropsychiatric disorders including addiction.^{84,85}

We found the caudate, NAC, SNr VeP, and STH revealed thalamic disconnections in DOC, consistent with the 'cortico–basal ganglia–thalamic loop's role in cognition.^{4,46,86–91} FC gains with frontal cortex for NAC, VeP, and Pu (including anterior cingulate cortex) align with the first two

nuclei driving dopaminergic midbrain activity^{29,60,61} and the latter's modulation by dopamine.^{29,60,61} Nuclei whose FC changes correlated with neurotransmitter system targets also coupled with lower-order networks including VN, AUDN (Ca and SNr) and CN (only SNr), and higher-order networks such as ECN (Ca and SNr) and FPN (only SNr). These findings complement previous work on network-specific subcortical organisation which identified the convergence of cingulo-opercular control and sensorimotor networks in the intermediate thalamus ("motor integration zones"), DMN and ECN in the caudate, and dorsal attention and VN in the pulvinar nucleus (thalamus),⁸⁶ perhaps indicating more specific targets for deep brain stimulation in DOC.

Our findings complement the 'mesocircuit model', which provides the rationale behind using dopaminergic agents post brain-injury,⁹² whereby widespread deafferentation leads to diminished central thalamus and frontal cortex projections, reducing afferent drive on striatal dopaminergic neurons and consequently failing to inhibit GPi.^{92,93} DAT dysregulation is associated with midbrain dopamine neuronal loss in mice⁹⁴ and abnormal dopamine clearance in humans.^{95,96} Furthermore, exogenous dopaminergic agonists (amantadine,^{97,98} methylphenidate,^{99,100} levodopa,^{101,102} bromocriptine,¹⁰³ and apomorphine)^{12,104–107} could theoretically increase firing rate of striatal neurons and restore loops between cortex, striatum, GPi, and central thalamic nuclei.¹⁰⁴ Methylphenidate blocks DAT presynaptically, elevating dopamine in the synaptic cleft;⁹³ though in posttraumatic MCS, presynaptic dopamine deficits in the striatum and central thalamus did not respond to DAT blockade but were reversed by exogenous administration of L-DOPA, restoring dopamine synthesis from the substantia nigra and VTA.¹⁰⁸ Diffuse axonal injury and dopaminergic vulnerability likely underpin executive dysfunction and altered cortico-striatal connectivity in TBI,¹⁰⁹ and moderate-severe TBI with cognitive impairments (albeit without DOC) showed dopamine dysfunction (reduced DAT in the caudate), with nigrostriatal tract and midbrain damage.¹¹⁰ Moreover, amantadine is the only recommended pharmacotherapy for DOC patients 4-16 weeks post-injury^{97,104,107} based on a randomised-controlled trial,^{98,111} likely driving frontostriatal circuits via glutamergic neuron activation,¹¹² which reinforces the need to validate neuromodulatory targets for optimising treatments.

At the macro-scale, we found multiple nuclei exhibited widespread FC alterations with the rest of the brain in DOC, supporting the 'cortico-basal ganglia-thalamo-cortical loop' framework of consciousness.¹¹³ Additional studies have investigated the specialised role of the GPe in supporting behavioural responsiveness and consciousness,^{8,9} and outlined thalamic and extra-thalamic

pallido-cortical pathways in healthy humans.¹¹⁴ Furthermore, primate work showed that ventral-anterior and ventral-lateral thalamic nuclei project directly to the basal ganglia, notably the caudate, and frontal cortical areas.^{115,116} These nuclei of the thalamo-basal ganglia circuitry play an important role in patients, as ventral-latero-ventral thalamic nuclei have been recently demonstrated to play a specialised role in DOC.¹¹⁷ Furthermore, DTI revealed abnormal structural connectivity between the caudate/putamen/pallidus, thalamus, and frontal cortex in DOC patients,¹¹⁸ and, cortico-striatal FC disruptions (caudate-anterior cingulate cortex) correlated with cognitive impairments in TBI, owing to downstream effects of axonal injury.¹¹⁹ Persistent neuronal dysfunction in DOC reflected restricted neocortical pyramidal neurons in FPN and altered thalamus and basal ganglia interactions.¹² In a patient 15 years post-TBI, vagal nerve stimulation increased PET metabolic activity in thalamus, basal ganglia, and cerebral cortex after 3 months.¹²⁰ Moreover, an independent study in 40 chronic DOC patients found globus pallidus atrophy associated with EEG frequency slowing and low behavioural arousal, implicating pallidal regulation of excitation-inhibition balance for complex cortical dynamics supporting consciousness, and suggesting it as a potential neuromodulatory target.¹²¹ While our study did not identify direct correlations with glutamatergic or GABAA receptors in the globus pallidus, both our findings and those of the EEG study underscore the relevance of the globus pallidus in disorders of consciousness, albeit through different methodological approaches.

Anaesthesia-related FC increases may reflect a compensatory mechanism to propofol's suppressive effects or mathematical artefacts caused by overall reduction of data variance. Hyperconnectivity occurs after acute chemogenetic inhibition of prefrontal cortex¹²² or primary somatosensory cortex in mice,¹²³ and as a common network response to neurological disruption in patients.^{124,125} Complexity-based measures (information-richness) may clarify mechanisms.²⁰ Conversely, DOC-related FC decreases may reflect permanent anatomical damage to pathways after brain injury, rather than transient neuromodulatory changes associated with pharmacological agents.¹²

Thus, pharmacological and pathological perturbations of consciousness (here indicated by loss of behavioural responsiveness), while broadly converging at cortical levels^{2,19,65,126} and in a brainstem study,¹⁷ diverge at the level of basal ganglia circuitry in FC and neurotransmitter dynamics. This has important translational implications, as DOC therapies may not be directly extrapolated from anaesthetic models at the level of basal ganglia, motivating the development of mechanistically

grounded state-specific models more accurately reflecting DOC neurobiology. While anaesthesia is widely used as a pharmacological, experimentally tractable model of the loss of behavioural responsiveness, it is important to acknowledge the differences between anaesthesia and DOC. First, pharmacological depression of otherwise intact neural networks and centres regulating physiological arousal cannot fully replicate the structural and functional damage present in disease-induced DOC. Second, most anaesthetics (including propofol) suppress not only behavioural responsiveness (and presumably consciousness, at least at high doses) but also physiological arousal. In contrast, some DOC patients, particularly those in UWS or MCS, may exhibit fluctuations in physiological arousal, such as cycles of eye opening and closing, despite limited or inconsistent behavioural responsiveness. Our results may therefore speak to the role of basal ganglia in both the loss of behavioural responsiveness observed in anaesthesia and DOC, but also the potential differences in physiological arousal between the two conditions.

Diagnostically, the boundary between fully ‘conscious’ and ‘unconscious’ states is blurred, motivating our primary focus on behavioural responsiveness as a more clear-cut outcome measure. Profound motor impairments can confound bedside assessments and lead to misdiagnosis in 30-40% of cases, for MCS, emergence from MCS, and locked-in syndrome.^{45,127,128} Functional neuroimaging reveals covert signs of preserved cognitive function through modulated cortical activity,^{43,45,129} where response to mental imagery tasks is indicative of awareness.¹³ Patients demonstrating preserved cognition without behavioural responsiveness exhibit ‘cognitive motor dissociation’,¹³⁰ and approximately 25% of patients without observable responses to commands perform cognitive tasks on fMRI/EEG;⁴⁴ though command-responsiveness may not be entirely correlated with connected consciousness (self-awareness), and unresponsiveness to command is not sufficient to infer absence of consciousness.¹³¹ We adopted an inclusive approach that acknowledges diagnostic uncertainty and clinical heterogeneity; our findings aligned with established behavioural stratifications and revealed meaningful DOC subgroup differences, supporting their internal validity and robustness.

Here, we not only provide a differential picture between pharmacological and pathological perturbations of responsiveness with relevance to basal ganglia, at FC and neuromodulatory hierarchies, but by stratifying patients by mental imagery task responsiveness, we show a more

1 refined picture of basal ganglia contributions to DOC. By presenting the significant correlations
 2 for each nucleus between cortical FC maps of the DOC subgroup comparison (contrast of *fMRI-*
 3 *versus fMRI+*) and neurotransmitter systems, that exhibited overlapping significance in the same
 4 direction as the results obtained from comparing *DOC versus two control groups*, we identified
 5 significant correlations between the caudate and 5-HTT, DAT, and GABA_A, and between the SNc
 6 and CB1 (Supplementary Fig. 3). This more fine-grained analysis allows for the identification of
 7 combinations of nuclei and neurotransmitter system maps whose relationship discriminates not
 8 only between behaviourally responsive and unresponsive individuals, but also between subgroups
 9 of DOC patients who are all behaviourally unresponsive, but some of whom provide evidence of
 10 covert consciousness.

11 Finally, the largest between-group difference in dFC in the DOC subgroup comparison was
 12 observed for the substantia nigra, a critical dopamine-producing region implicated in motor
 13 control.⁵⁸ SNc releases dopamine to striatal neurons projecting to basal ganglia, which in turn is
 14 connected to the thalamus and motor cortex,^{58,132} whereas SNr receives projections from dopamine
 15 neurons in the SNc.^{133,134} Crucially, the substantia nigra is not only responsible for FC changes in
 16 pathological loss of responsiveness overall, but is even more prominent in the cohort of more
 17 unresponsive DOC patients unable to demonstrate command-following during fMRI tasks. This
 18 suggests that substantia nigra connectivity may serve as a marker specific to DOC patients,
 19 particularly those at the more unresponsive end of the spectrum, further reinforcing the
 20 ‘mesocircuit model’⁹² where dopaminergic neuronal cell bodies in substantia nigra (SNc) and VTA
 21 project to frontal cortex and striatum through multiple dopaminergic pathways, including
 22 nigrostriatal, mesolimbic, and mesocortical.⁹³ This DOC specificity indicates that certain patients
 23 may require alternative routes for neuromodulatory interventions, encouraging future studies on
 24 personalised functional connectome mapping.¹⁰⁶ Beyond pharmacotherapy, ideal targets for brain
 25 stimulation remain unsettled.¹³⁵ In 40 patients (MCS/UWS), DBS of the centromedian-
 26 parafascicular thalamic complex was most effective when extended to parafascicular and
 27 subparafascicular nuclei and white matter projection pathways of default ascending arousal
 28 network, hypothalamus, brainstem and medial frontal cortex.²² Intact striatal grey matter correlated
 29 with consciousness improvement,²² supporting a striatal “gating role” in restoring mesocircuit
 30 activity,¹³⁶ complementing our findings. Furthermore, multitarget stimulation has been shown to
 31 improve gait in Parkinson’s (substantia nigra-STH)¹³⁷ and seizures in drug-resistant epilepsy

(anterior-pulvinar thalamic nuclei).¹³⁸ As consciousness models mature, patient care may pivot towards combination therapies i.e. multi-drug therapy and multi-site cortical-subcortical stimulation, both functional connectome- and receptor-informed,¹⁰⁶ to guide treatment and reduce prognostic uncertainty.

Our study has several limitations. Firstly, basal ganglia subdivisions were not segmented at the basis of individual subjects in their native brain space (the processing pipeline we adopted is standard in neuroimaging studies), however, this most recently developed high-resolution atlas should be able to tolerate the normalisation process, as it was validated following an assessment of intra- and inter-rater labelling reliability.²⁹ Secondly, we acknowledge the relatively small sample cohort sizes, and patient heterogeneity in diagnosis and aetiology; though, patient selection was restricted by strict exclusion criteria to ensure accuracy of subcortical analyses. Consequently, this constrains interpretability about ‘true unconsciousness’ and generalisability across the broader spectrum of DOC, which is why we focused specifically on behavioural unresponsiveness, which is shared by DOC and anaesthesia despite the differences that also exist between these two conditions. However, such criteria are not unique to our study but are adopted in previous publications with these datasets,^{17,19} and our exclusion rate aligns with recent broader DOC literature.²² Nonetheless, the rarity of DOC patients with sufficiently analysable imaging data underscores the field’s challenges, and future studies will be important for validation and extension of our findings.

Data availability

Access to DOC patient data is limited to qualified researchers for non-commercial use, in order to protect patient confidentiality. The UK Health Research Authority assigns responsibility for safeguarding the data to the Chief Investigators of the original studies (Dr. Judith Allanson and Prof. David Menon, or anyone else to whom responsibility is given). To request access, please contact the Data Access Committee: Dr. Judith Allanson (judith.allanson.1@gmail.com), Prof. David Menon (dkm13@cam.ac.uk), or Dr. Emmanuel Stamatakis (eas46@cam.ac.uk). The Western dataset is available on the OpenNeuro data repository (doi:

10.18112/openneuro.ds003171.v2.0.1). The Cambridge dataset is available on request from Dr. Emmanuel Stamatakis (eas46@cam.ac.uk) or Dr. Ram Adapa (ra342@cam.ac.uk).

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Competing interests

The authors declare no competing interests.

Supplementary material

Supplementary material is available at *Brain* online.

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6 Figure legends

7 **Figure 1 Whole-brain resting-state (rs) functional connectivity (FC) maps in healthy subjects**
8 **in pharmacological (propofol-induced) loss of consciousness.** Results are computed from seed-
9 to-voxel rs-FC analysis across respective basal ganglia nuclei [bilateral] seeds — putamen (Pu),
10 caudate (Ca), nucleus accumbens (NAC), globus pallidus externus (GPe), globus pallidus internus
11 (GPi), substantia nigra pars compacta (SNc), substantia nigra pars reticulata (SNr), ventral pallidus
12 (VeP), and subthalamic nucleus (STH) — and the rest of the brain (A–I). Colour bar denotes the
13 strength of the t-statistic. Unthresholded cortical SPM-t maps (A–I, *left* column) and thresholded
14 cortical SPM-t maps (A–I, *middle* column) of the deep sedation vs. awake contrast are shown.
15 Radar plots indicate the intrinsic connectivity network (ICN) spatial involvement (ICNi) of brain
16 regions in the deep sedation vs. awake contrast i.e. shows the voxel overlap between the FC results
17 and canonical ICNs (A–I, *right* column). These canonical ICNs were defined by an atlas³⁵
18 containing 10 well-matched resting-state networks (RSN) from the *ICN_atlas* toolbox
19 (https://www.nitrc.org/projects/icn_atlas/).³⁶ Description of ICN-RSN atlas: VN 1, 2, and 3
20 (“*visual network*”): medial, occipital pole, and lateral visual areas; DMN (“*default mode*
21 *network*”): medial parietal (precuneus and posterior cingulate), bilateral inferior–lateral–parietal,
22 and ventromedial frontal cortex; CN (“*cerebellum network*”): cerebellum; SMN (“*sensorimotor*
23 *network*”): supplementary motor area, sensorimotor cortex, and secondary somatosensory cortex;
24 AUDN (“*auditory network*”): superior temporal gyrus, Heschl's gyrus, and posterior insular, and
25 includes primary and association auditory cortices; ECN (“*executive control network*”): medial–
26 frontal areas, including anterior cingulate and paracingulate; FPN1 and 2 (“*frontoparietal*
27 *network*”): frontoparietal areas; these are the only maps to be strongly lateralised. In addition,
28 FPN1 corresponds strongly to perception–somesthesia–pain, and FPN2 to cognition–language
29 paradigms, consistent with Broca's and Wernicke's areas.^{35–37}

Figure 2 Significant correlations of 19 major PET receptors and transporters and cortical SPM-*t* maps of deep vs. awake contrast (cortical maps of changes in functional connectivity).

Red represents significant positive correlations and blue represents the significant negative correlations.

Figure 3 Whole-brain resting-state (rs) functional connectivity (FC) maps in DOC patients in pathological (DOC-induced) loss of consciousness. Results are computed from seed-to-voxel

rs-FC analysis across respective basal ganglia nuclei [bilateral] seeds — putamen (Pu), caudate (Ca), nucleus accumbens (NAC), globus pallidus externus (GPe), globus pallidus internus (GPi), substantia nigra pars compacta (SNc), substantia nigra pars reticulata (SNr), ventral pallidus (VeP), and subthalamic nucleus (STH) — and the rest of the brain (A–I). Colour bar denotes the strength of the *t*-statistic. Unthresholded cortical SPM-*t* maps (A–I, *left* column) and thresholded cortical SPM-*t* maps (A–I, *middle* column) of the DOC vs. control contrast are shown. Radar plots indicate the intrinsic connectivity network (ICN) spatial involvement (ICNi) of brain regions in the DOC vs. control contrast i.e. shows the voxel overlap between the FC results and canonical ICNs (A–I, *right* column). These canonical ICNs were defined by an atlas³⁵ containing 10 well-matched resting-state networks (RSN) from the *ICN_atlas* toolbox (https://www.nitrc.org/projects/icn_atlas/).³⁶ Description of ICN-RSN atlas: VN 1, 2, and 3 (“*visual network*”): medial, occipital pole, and lateral visual areas; DMN (“*default mode network*”): medial parietal (precuneus and posterior cingulate), bilateral inferior–lateral–parietal, and ventromedial frontal cortex; CN (“*cerebellum network*”): cerebellum; SMN (“*sensorimotor network*”): supplementary motor area, sensorimotor cortex, and secondary somatosensory cortex; AUDN (“*auditory network*”): superior temporal gyrus, Heschl's gyrus, and posterior insular, and includes primary and association auditory cortices; ECN (“*executive control network*”): medial–frontal areas, including anterior cingulate and paracingulate; FPN1 and 2 (“*frontoparietal network*”): frontoparietal areas; these are the only maps to be strongly lateralised. In addition, FPN1 corresponds strongly to perception–somesthesia–pain, and FPN2 to cognition–language paradigms, consistent with Broca's and Wernicke's areas.^{35–37}

Figure 4 Significant correlations of 19 major PET receptors and transporters and cortical SPM-*t* maps for both “Western” controls and “Cambridge” controls. The correlations (Fig. 4) and spin-tests (Supplementary Fig. 2) are plotted for each nucleus between the “Western” controls and “Cambridge” controls cortical maps of changes in seed-based functional connectivity, for which correlations are significant after accounting for spatial autocorrelation ($p_{spin} < 0.05$), in both comparisons (DOC versus “Western” controls, and DOC versus “Cambridge” controls) and with the same sign red represents significant positive correlations and blue represents the significant negative correlations. The magnitude reported is the smaller of the two observed correlation magnitudes, representing the more conservative estimate of the effect.”

Figure 5 Whole-brain resting-state (rs) functional connectivity (FC) maps *fMRI*- “task non responders” versus *fMRI*+ “task responders”. Results are computed from seed-to-voxel rs-FC analysis across respective basal ganglia nuclei [bilateral] seeds — putamen (Pu), caudate (Ca), nucleus accumbens (NAC), globus pallidus externus (Gpe), globus pallidus internus (GPi), substantia nigra pars compacta (SNc), substantia nigra pars reticulata (SNr), ventral pallidus (VeP), and subthalamic nucleus (STH) — and the rest of the brain. Colour bar denotes the strength of the *t*-statistic. Unthresholded cortical SPM-*t* maps (A–I, left column) and thresholded *t*-maps (A–I, right column) of the *fMRI*- versus *fMRI*+ contrast are shown.

Figure 6 Behavioural task in patients with disorders of consciousness. Plots display dFC scores for each DOC subgroup (i) *fMRI* negative/task non-responders ($n=15$) and (ii) *fMRI* positive/task responders ($n=7$), where dFC is the standardised mean absolute voxel-wise difference between each patient’s individual FC map and the DOC subgroup contrast (*fMRI*- versus *fMRI*+). Lower dFC indicates closer alignment to the DOC subgroup contrast, whereas higher dFC indicates greater deviation. Boxes depict the median and interquartile range (IQR). Statistical comparisons between groups were performed for each nucleus using FDR-corrected Welch’s *t*-tests and Wilcoxon rank-sum tests. Among all nuclei, the largest between-group difference in dFC was observed for the substantia nigra pars compacta (SNc).

Table 1 Demographics for DOC patients

Patient	Sex	Age, years	Aetiology	Months Post-Injury	Diagnosis (MCS or UWS)	CRS-R	Tennis Task
1	Male	21	TBI	45	MCS+	11	Positive
2	Male	57	TBI	14	MCS-	12	Negative
3	Male	47	TBI	4	MCS+	10	Negative
4	Male	36	TBI	34	UWS	8	Negative
5	Male	17	Anoxic	46	UWS	11	Negative
6	Female	38	Anoxic	9	MCS-	10	Negative
7	Female	38	TBI	13	MCS+	11	Positive
8	Male	29	TBI	68	MCS+	10	Positive
9	Male	23	TBI	4	MCS+	7	Positive
10	Female	70	Cerebral bleed	11	MCS+	9	Negative
11	Female	30	Anoxic	6	MCS-	9	Positive
12	Female	34	Anoxic	6	UWS	8	Negative
13	Male	22	Anoxic	5	UWS	7	Negative
14	Male	37	Anoxic	14	UWS	7	Negative
15	Female	62	Anoxic	7	UWS	7	Negative
16	Male	46	Anoxic	10	UWS	5	Negative
17	Male	21	TBI	7	MCS+	11	Negative
18	Male	67	TBI	14	MCS-	11	Positive
19	Female	55	Hypoxic	Unknown	UWS	12	Negative
20	Male	28	TBI	Unknown	MCS+	8	Positive
21	Male	22	TBI	Unknown	MCS+	10	Negative
22	Female	28	ADEM	Unknown	UWS	6	Negative

CRS-R = Coma Recovery Scale-Revised; MCS = minimally conscious state; UWS = unresponsive wakefulness syndrome.

1 **Table 2 Demographics for Control Dataset2 ('Cambridge')**

Healthy volunteer	Sex	Age, years
1	Male	32
2	Female	29
3	Female	39
4	Male	27
5	Male	36
6	Female	43
7	Female	46
8	Female	42
9	Male	19
10	Female	37
11	Male	25
12	Male	52
13	Male	32
14	Female	23
15	Female	46
16	Female	33
17	Female	31
18	Male	42
19	Female	49
20	Male	28
21	Male	23
22	Female	36
23	Female	43
24	Male	25
25	Female	29

2
3

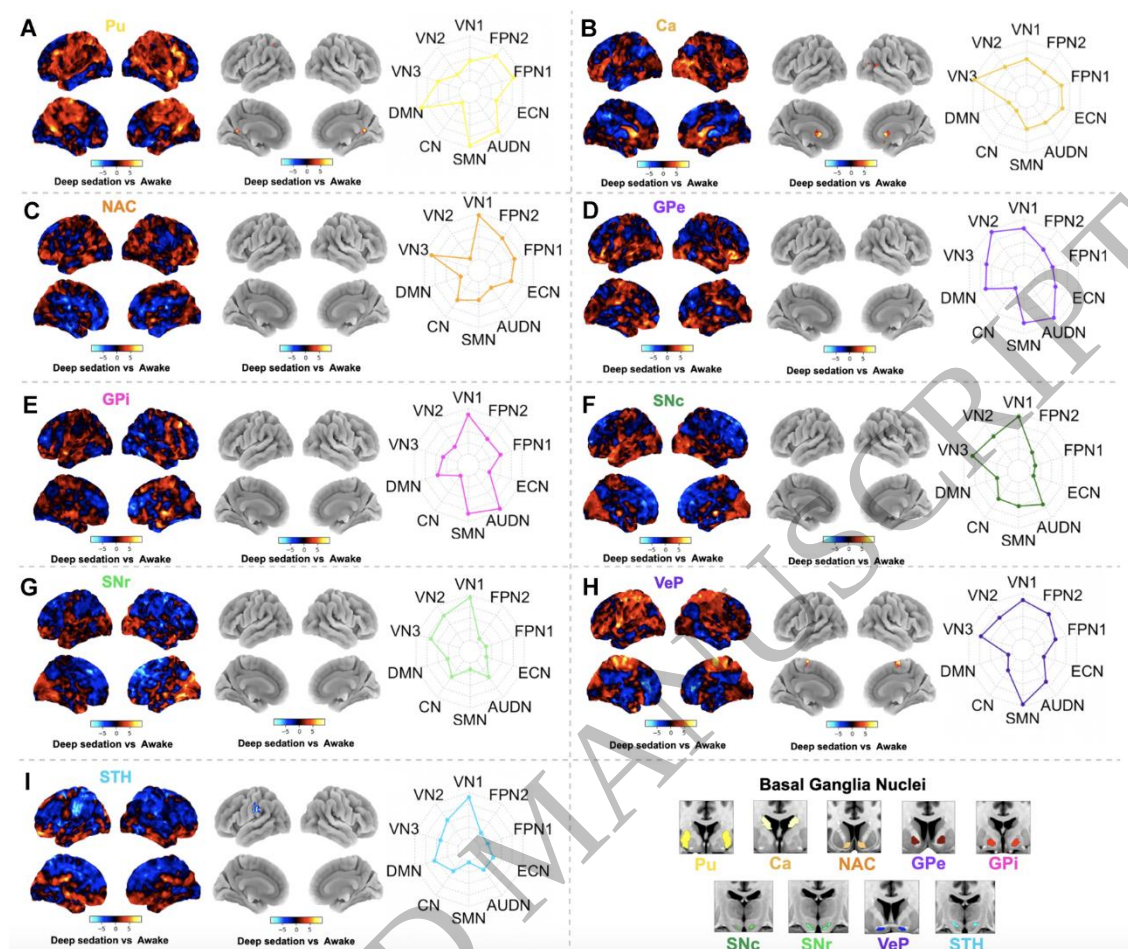


Figure 1
150x125 mm (x DPI)

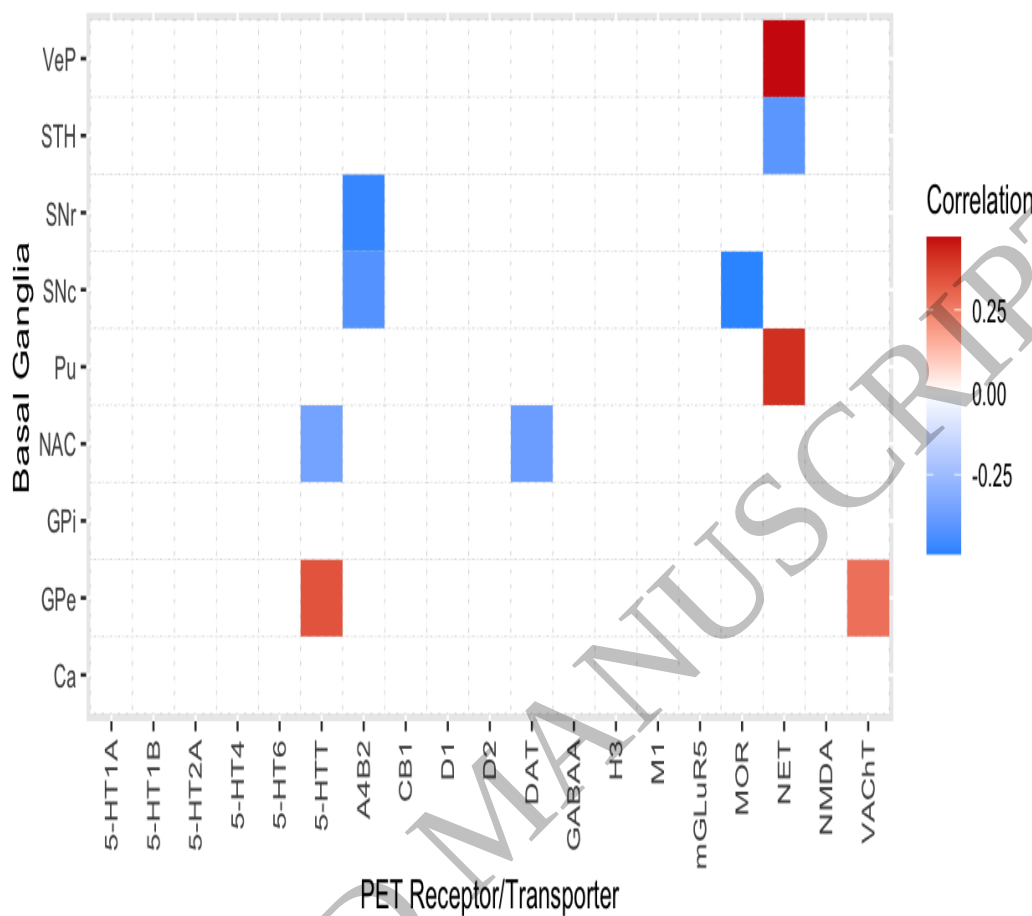


Figure 2
439x220 mm (x DPI)

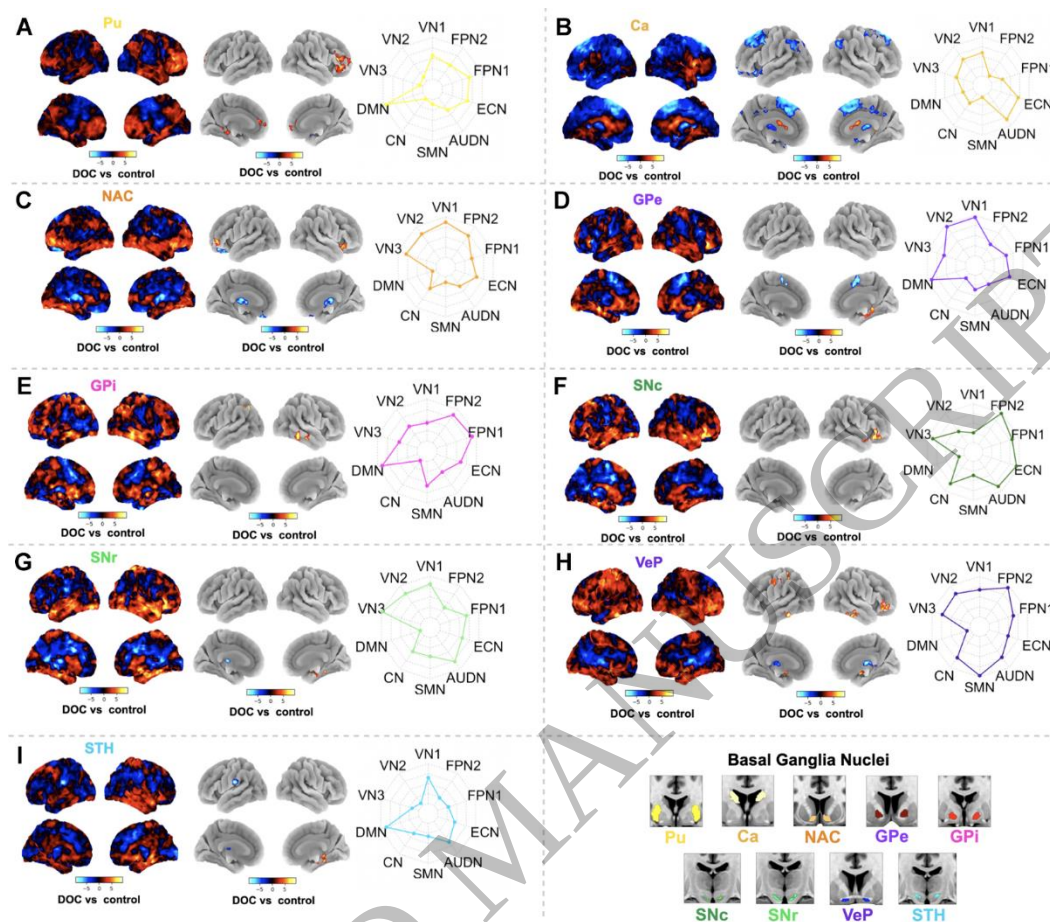


Figure 3
140x121 mm (x DPI)

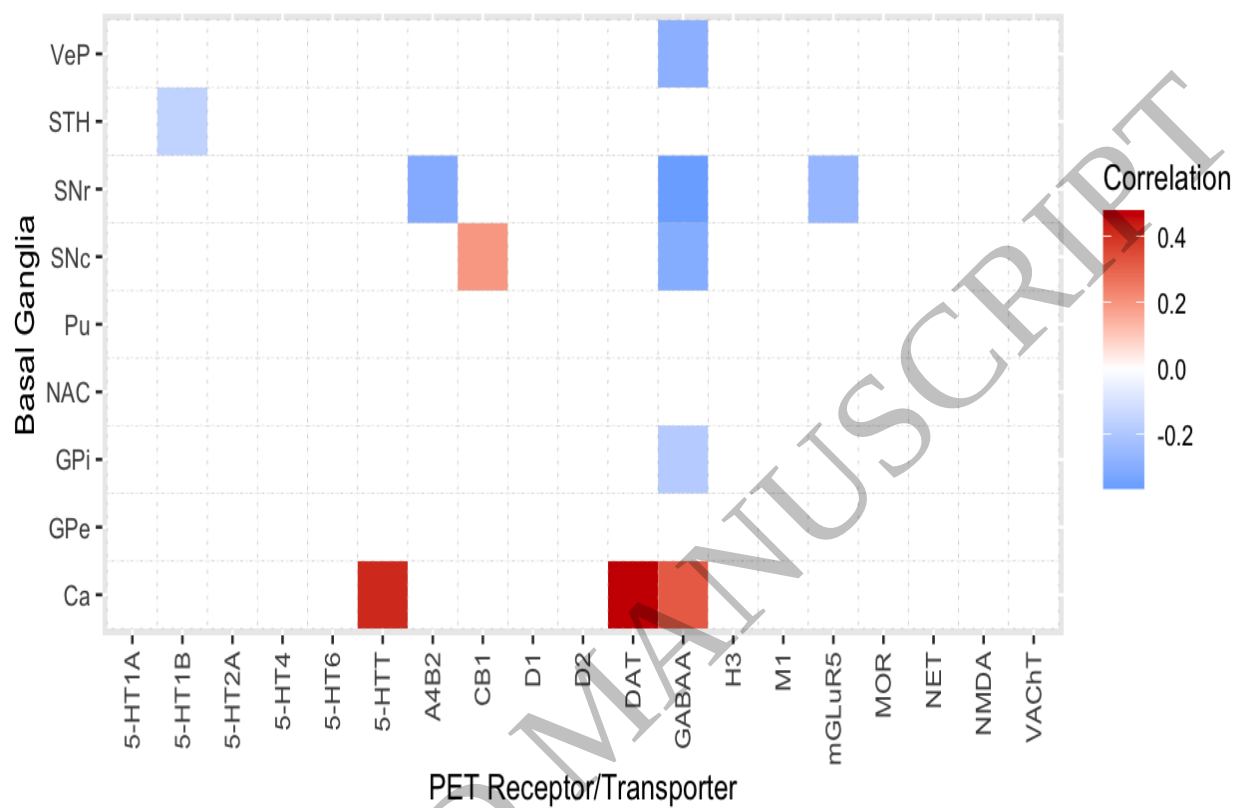


Figure 4
333x206 mm (x DPI)

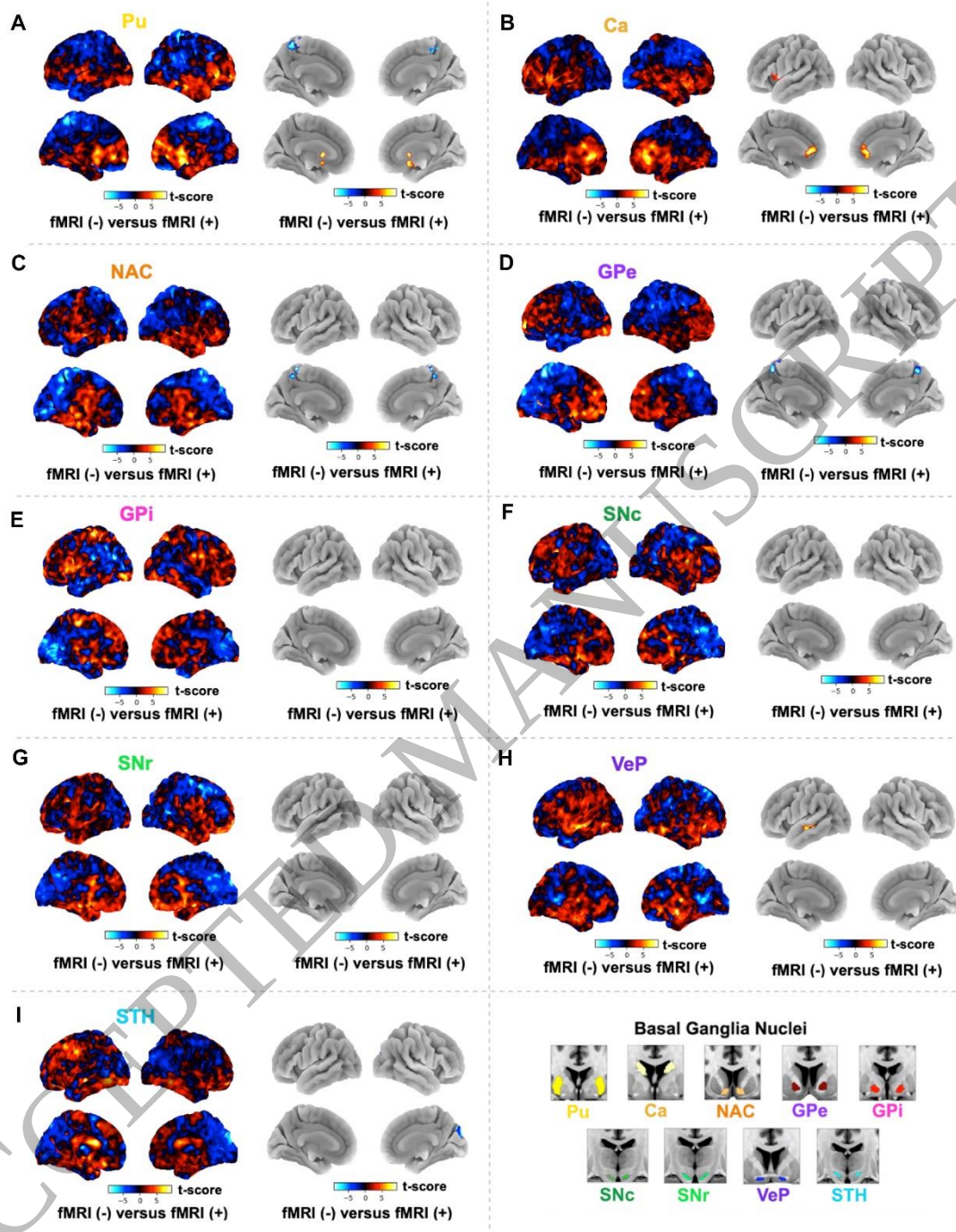


Figure 5
146x183 mm (x DPI)

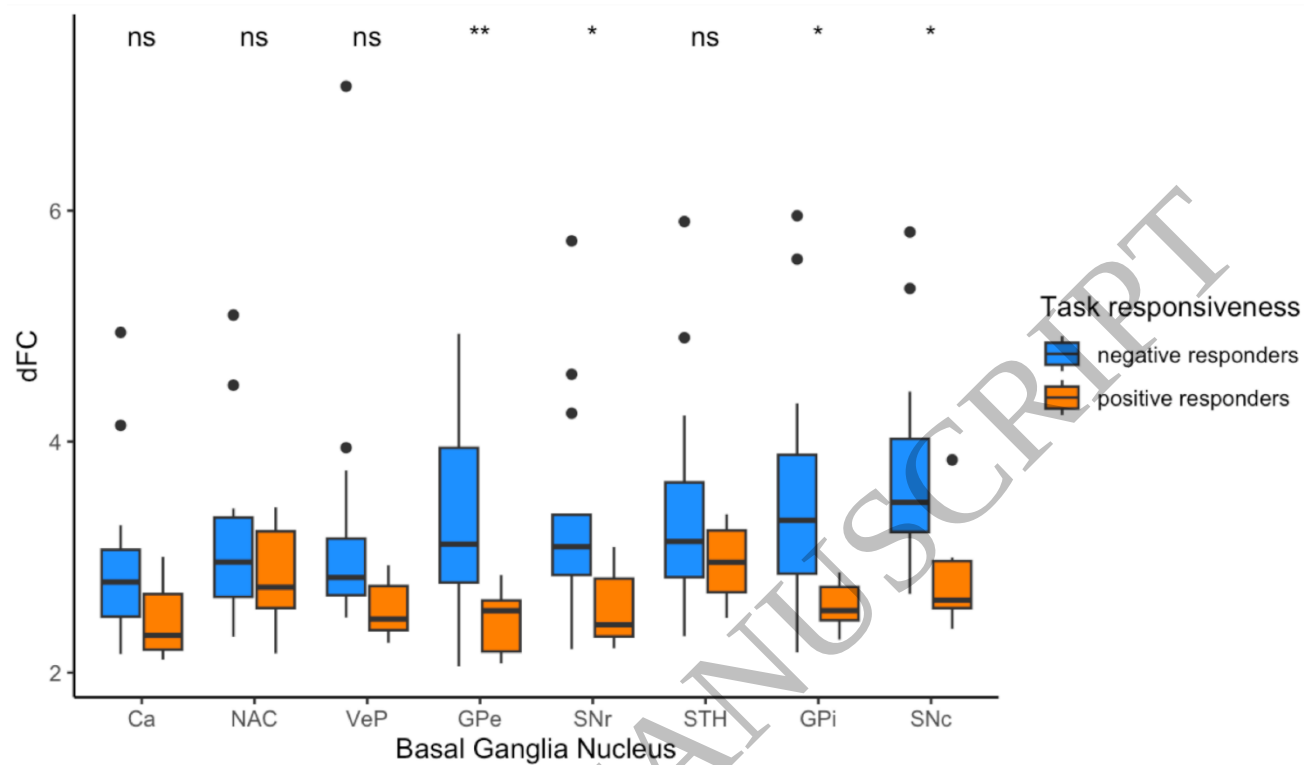


Figure 6
173x102 mm (x DPI)